Cooperative Connection Transformer for Remote Sensing Image Captioning

Kai Zhao and Wei Xiong

***Abstract*— Feature extraction is fundamental for successful remote sensing image captioning (RSIC). The representation forms of grid features and region features differ significantly. Grid features can provide fine-grained information while retaining background information for subsequent processes but lack salient target information. Region features can provide object-level information but lack contextual information. Thus, to fully utilize the advantages of both and achieve RSIC tasks, we propose a cooperative connection Transformer (CCT). In the encoder, we specifically designed grid and region features. We used the features after global average pooling (GAP) as grid features to provide global background information. We also extract region features to provide salient region information. To solve the information loss caused by region feature extraction, we propose mapping region features and fusing them with grid features using a fusion attention module that filters out the information redundancy and noise generated by fusion. Cooperative connection attention is added to the decoder to better utilize different feature types to improve its perception of different features by directly connecting them to each decoder layer. To address the lack of region feature annotations in the RSIC field, we provide region feature annotations based on published datasets and extract region features accordingly. Extensive experiments demonstrate the effectiveness and superiority of the proposed method. Labeled region annotations are available at https://github.com/zk-1019/region-annotations.**

***Index Terms*— Image processing, remote sensing image captioning (RSIC), Transformer.**

I. Introduction

With the development of deep neural networks (DNNs), remote sensing images have undergone significant progress in computer vision tasks such as object detection [1], [2], [3] and image segmentation [4], [5]. Inspired by advancements in single modalities, remote sensing imaging research has changed from single modality to multimodality [6]. In the field of multimodality, research on vision-language tasks, such as visual question answering (VQA) [7], [8], [9] and image captioning [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], has achieved specific results.

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As a cross-modal task combining computer vision and natural language processing, remote sensing image captioning (RSIC) aims to generate sentences that describe the content of a given remote sensing image. Since Qu et al. [14] first proposed the RSIC task, extensive research has been conducted on it. RSIC research has gone through the following stages: from traditional to encoder–decoder-based methods. Traditional methods include retrieval [32] and template-based [33] methods; however, these methods cannot generate flexible sentences, and content matching between text and images is unsatisfactory. Thus, Qu et al. [14] used convolutional neural networks (CNNs) as encoders to extract image features and a recurrent neural network (RNN) [34] as a decoder to generate captions, which determined the basic framework of the RSIC, and the current RSIC models all of which followed the encoder–decoder framework [11], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [35].

Image and text matching has always been a complex problem, and as different modal features exist, image feature always represents as $H \times W$ dimensional feature maps with C channels. Text features are represented as tokenized vectors. Researchers have made tremendous efforts to achieve a better match between these two [17], [18], [19], [22], [27], [29]. In the early stages, handcrafted features [36], [37] were used to represent remote sensing images; however, their performance was unsatisfactory. Therefore, CNN-based encoders have become the mainstream method for feature extraction. However, the features extracted by CNNs cannot cause the decoder to focus on a critical region. Attention mechanisms significantly promote matching between the two modalities, which causes the decoder to attend to essential regions. Lu et al. [38] first introduced attention mechanisms into RSIC and achieved significant success. Since then, the attention mechanism has become dominant in RSIC. To match remote sensing images and texts better, researchers have introduced additional tasks or attributes to guide text generation [20], [21], [22], [23], [24], [25].

Image feature extraction is the foundation of RSIC tasks, significantly determining the quality of generated sentences. Grid and region features, as two different representation forms, are shown in Fig. 1. Grid features usually refer to the features extracted by CNNs. The extracted feature maps are divided into $H \times W$ grid regions in spatial dimensions. As the commonly used feature, grid features can provide sufficient fine-grained information while providing global information that could be useful in the subsequent process. Therefore,

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Fig. 1. (a) Grid features, the extracted feature maps are split into equal-sized grids. (b) Region features, represented by sets of salient targets or regions.

many researchers tried to represent the remote sensing image more efficiently based on the grid feature [19], [20], [21], [25]. However, grid features lack object-level information, where the features are split into equal-sized grids, unable to provide salient object information.

Region features were initially used for object-detection tasks. Bottom-up attention [39] proves that region features work well in improving cross-modal tasks and gradually dominate natural image captioning tasks. Region features are a set of vectors, with each vector representing a salient object or region, thereby providing object-level information.

Remote sensing images usually contain numerous targets that typically occupy a small area, making them difficult to distinguish from the background. The object-level information provided by region features is helpful for the RSIC, which can help the model better define the focus of a description. However, most of the proposed RSIC models are based on grid features, and there has been no research on region features in the RSIC. It remains uncertain whether region features that perform well in natural image captioning can be used in RSIC.

It is more difficult to use region features for RSIC than grid features. One challenge was the lack of relevant annotations. In natural image captioning, researchers rely on Microsoft COCO (MS COCO) [40] dataset, including caption annotations and object-instance (bounding box) annotations. Therefore, extracting region features and applying them to image caption tasks is easy. There are limited datasets in the RSIC that contain only caption annotations without corresponding region annotations. The extraction of region features requires object detectors, and the training of these detectors requires region annotations to perform regression on the candidate anchors generated by the region proposal network (RPN) with ground-truth bounding boxes.

To utilize region features, we first addressed the problem of the lack of region annotations. Based on the public RSIC datasets, we uniformly labeled region annotations, with more than 39 400 object-instance bounding boxes on 13 634 remote sensing images, and extracted region features based on those annotations. We extensively experimentally validate the feasibility of using region features in RSIC tasks. We also propose a cooperative connection Transformer (CCT) to utilize region and grid features better.

Therefore, the main contributions of this article are as follows.

1) We propose a CCT to utilize the region and grid features jointly. The performance of the model is demonstrated through extensive experiments.
2) We propose a mapping region feature and a fusion attention module to utilize the region features better. The mapping region features compensate for the information loss caused by region feature extraction, whereas the fusion attention module filters out the redundancy and noise caused by feature fusion.
3) We propose cooperative connection attention to enhance the decoder's perception of different features. The decoder is forced to focus on the global background and local salient information by directly connecting different features to the decoder and weighted cross-attention.
4) To the best of our knowledge, this is the first study that applies region features to an RSIC task. We uniformly labeled the region annotations of the UCM-captions, Sydney-captions, and RSICD, providing a novel feature-utilization method for RSIC.

II. Related Works

A. Encoder–Decoder Model

Since the first encoder–decoder model was proposed, encoder–decoder-based models have dominated RSIC tasks. In subsequent research, various improvements have been made to the basic encoder–decoder framework to meet the requirements of RSIC tasks and improve the quality of the generated description text. Those research mainly focuses on three parts: exploring better encoder, exploring better attention mechanism, and exploring better decoder.

1) Exploring Better Encoder: The aim is to obtain better image feature representations using different encoder methods. In [16], a layered hierarchical rearrangement multilayer perceptron (HMLP) model was proposed, with the encoder being a layered rearrangement multilayer perceptron to improve the object recognition ability. In [17], a joint-training two-stage (JTTS) model was proposed, in which the encoder and decoder can be trained jointly through joint loss. In [18], a dual feature enhancement network (DFEN) containing an image enhancement module was proposed to obtain adequate ground object information. In [19], a multiscale multi-interaction network (MMN) was proposed with a multi-interaction feature representation module to improve feature interaction.

2) Exploring Better Attention Mechanism: The aim is to improve the cross-modal matching. In [20], a multisource interactive stair attention (MSISA) was proposed that utilizes sentences and visual feature separately. In [21], structured attention (SA) was proposed to utilize the structured characteristics of the semantic content of remotely sensed images. In [22], a recurrent attention (RA) mechanism was proposed to extract high-level attention maps from encoded and nonvisual features to help the decoder identify effective information. In [23], the global visual feature-guided attention (GVFGA), which uses fused image features and a linguistic state to guide the caption generation process,

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was proposed. In [24], a visual-semantic co-attention (VSCA) was proposed for grain concepts and regions for cross-modal interaction. In [25], an attention-based global–local captioning model (GLCM) was proposed to capture global–local features.

3) Exploring Better Decoder: The aim is to generate a more accurate descriptive text. In [26], a support vector machine (SVM) decoder was proposed to generate captions at a low cost. In [22], a semantic gate was proposed to incorporate the hidden states of long short-term memory (LSTM) [41]. In [23], a linguistic state was proposed to help LSTM guide the visual–textual attention process.

B. Transformer

Due to the excellent sequence processing performance, Transformer [42] has been successfully used in the cross-modal field. Therefore, Transformer is one of the most commonly used RSIC models. In [27], a novel deformable Transformer (DFT) structure utilizing a variable-point multiplication attention mechanism was proposed to learn the multiscale features of remote sensing images from the foreground and background information. In [28], a novel Transformer structure was proposed, with a dropout layer, a residual connection, and adaptive feature fusion on top of the original Transformer. Reinforcement learning has also been used to improve the models' performance. In [29], a grid feature Transformer (GFT) was proposed, in which grid features were used to provide visual features, and the Transformer outputs semantic captions. In [43], a bidirectional encoder representation of the Transformer was used to extract caption features, and it also built a strong correlation established between images and captions. In [30], a multiscale Transformer containing a controller for controlling the caption types as demanded was proposed.

Inspired by [31] and [42], our model is also based on the Transformer. Changes were made to the encoder (for feature extraction) and the decoder (for text generation) according to the characteristics of the remote sensing images.

III. Construction of Region Feature Annotations

Currently, the most commonly used datasets are the UCM-captions [14], Sydney-captions [14], and RSICD [38]. The UCM-captions contained 21 scene categories, including 2100 remote sensing images, each with an image of 256×256 pixels. The Sydney-captions contained a total of 613 remote sensing images in seven scene categories, each with an image of 500×500 pixels. The RSICD is currently the largest dataset in the RSIC, which contains 10 921 remote sensing images, each with 224×224 pixels, and each image is annotated with five description sentences. The data are organized into 30 categories, and there are overall 3323 words in the corpus. However, the datasets mentioned above only contain caption annotations and do not contain the corresponding region annotations. Therefore, to provide region features for extraction, we labeled the region annotations of the three datasets.

Owing to the large number of images and wide variety of categories in the three datasets, we propose annotation principles to ensure format consistency, maintain consistent annotation styles on different images, and provide the detailed and accurate region features. All labeling was performed under the guidance of the following principles.

1) *Content Consistency:* When an image is labeled, the annotation focus is determined by the corresponding caption content; for example, if the caption annotation mainly describes sparse residential areas, we focus on residential areas without labeling trees and other objects in the image. For example, in Fig. 2, we labeled only residential areas. Similarly, we ignore the surrounding buildings and other parts when labeling baseball fields.
2) *Category Consistency:* Annotation categories are determined consistently with the labeled dataset. At the same time, the frequency of words in the corpus is analyzed, the words are arranged according to their occurrence frequency, and additional object categories are added accordingly, such as "road" and "football field," as shown in Table I for specific categories.
3) *Object Independence:* Remote sensing images usually contain many objects, with same-category objects appearing more than once in a given image. Therefore, in the labeling process, different objects belonging to the same category that appear in the same image are labeled with separate bounding boxes. Targets of the same category can be concentrated in a particular area, making it difficult to distinguish them. Objects that are not easily distinguishable are labeled as entire object instances (bounding boxes). For example, in the airplane and tennis court category labeling shown in Fig. 2, each object was labeled with a separate bounding box, whereas objects in the industry were labeled as a whole for different areas. For irregularly shaped objects, labels may extensively cover the main part of the target, such as playgrounds, and we attempted to cover the main part when labeling it.

We labeled 13 634 remote sensing images from the UCM-captions, Sydney-captions, and RSICD, and 39 703 object instances according to the above principles. In UCM-captions, we defined 23 categories; in Sydney-captions, we defined 12 categories. In RSICD, we defined 36 categories, some of which are shown in Fig. 2. The number of annotation instances for each category is listed in Table I. Our annotation format is based on the MS COCO annotation format. All annotations were stored in JSON files.

IV. Approach

This section introduces the proposed CCT, which is formulated within the encoder–decoder framework. The CCT is shown in Fig. 3. When a remote sensing image is input into the model, it is first converted into a feature map $\boldsymbol{X}$ by DNNs, following which grid features $\boldsymbol{x}_g$ are obtained after global average pooling (GAP) [44]. Simultaneously, the region features $\boldsymbol{x}_r$ were extracted using a two-stage object detector. Thus, the mapping region features $\boldsymbol{x}_\Omega$ are obtained by masking

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TABLE I
CATEGORY OF DIFFERENT DATASETS

Dataset	Category	Annotations Per Category	Category	Annotations Per Category	Category	Annotations Per Category
UCM-captions	airport	266	freeway	363	storage tanks	199
	airplane	194	golf course	228	tennis court	271
	beach	101	harbor	416	overpass	196
	buildings	283	intersection	198	farmland	126
	chaparral	100	medium residential	463	parking lot	154
	dense residential	316	river	408	boats	1189
	baseball diamond	102	runway	181	mobile homes	588
	forest	293	sparse residential	106		
Sydney-captions	airport	49	residential	358	parking lot	64
	industrial	543	river	253	bushes	648
	meadow	406	runway	435	sea	93
	ocean	97	airplane	51	highway	18
RSICD	playground	1123	plane	1959	sparse residential	300
	buildings	4396	airport	576	stadium	290
	bridge	1210	port	549	football field	245
	river	2862	bareland	427	viaduct	435
	storage tanks	1966	beach	668	railway station	263
	road	557	pond	917	resort	294
	parking lot	938	center	270	school	299
	square	403	church	248	forest	301
	baseball field	635	commercial	174	industrial	644
	meadow	416	dense residential	1060	medium residential	332
	tennis court	223	desert	304	mountain	340
	basketball field	211	farmland	368	park	381

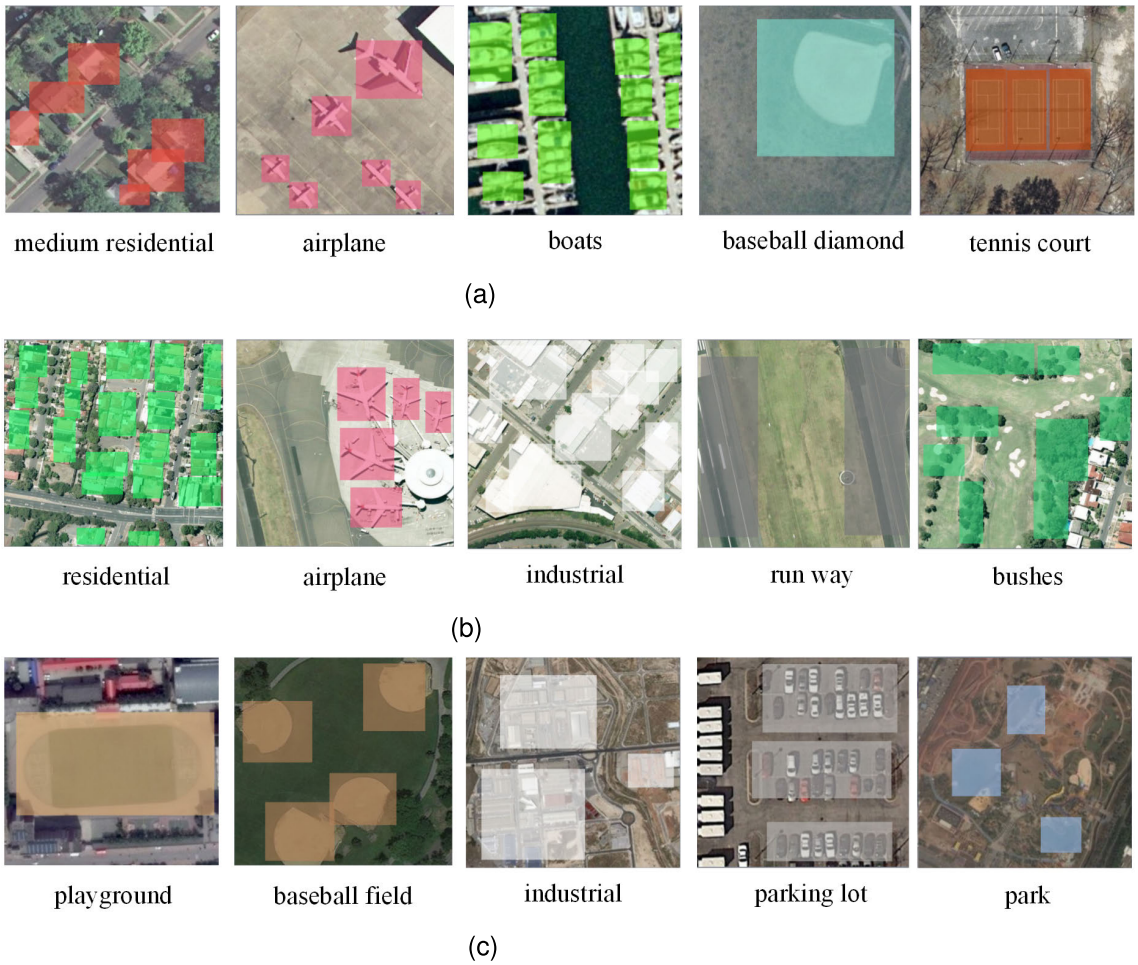


Fig. 2. Annotations in the datasets. (a) Annotation example in UCM-captions. (b) Annotation example in Sydney-captions. (c) Annotation example in RSICD. The different colored bounding boxes represent different categories.

the corresponding feature maps. After passing through a fusion attention module, the mapping region and grid features are fused to yield the fused features $\boldsymbol{x}_f$. In the decoder, the region features and fusion features obtained by the encoder are

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connected to each layer of the decoder, and attention weights are obtained by the cooperative connection attention to guide the text generation process, yielding predictions.

In Sections IV-A–IV-E, we discuss the extraction of grid and region features, obtained mapping region features, fusion attention module, and cooperative connection attention mechanism.

A. Grid Features

Region features provide salient object-related information but no global background-related information. Therefore, we used grid features to supply background information to the model. The grid features are extracted using a CNN. After the remote sensing image $\boldsymbol{I}$ was presented to the model, it was first converted into a feature map $\boldsymbol{X}$

$$\boldsymbol{X} = \mathrm{CNN}(\boldsymbol{I}). \tag{1}$$

Remote sensing images usually contain more content than natural images, and background information is essential for RSIC. Therefore, to obtain the background information, we used ResNet101 [45] as the backbone network to extract the grid features. ResNet can be divided into C1–C5 blocks. It is generally believed that features extracted from deeper layers contain more high-level semantic and global information. However, using the output of the fully connected layer as grid features will likely cause information loss because C5 has 2048 output channels. In contrast, the output of the fully connected layer is $1000 - d$, implying a loss of channel information. In addition, global features should be rotation- and translation-invariant. Therefore, we considered the output of the C5 module after the GAP as the grid feature as follows:

$$\boldsymbol{X} = \mathrm{ResNet}_{c5}(\boldsymbol{I}). \tag{2}$$

The kth element of the grid feature is

$$\boldsymbol{x}_g(k) = \mathrm{GAP}(\boldsymbol{X}) = \frac{1}{H \times W} \sum_{i=0}^{H} \sum_{j=0}^{W} \boldsymbol{X}_{[i,j,k]} \tag{3}$$

where H and W represent the length and width of the feature map respectively; thus, the global background information was obtained.

B. Region Features

The region representation of a remote sensing image can be construed as a set of K vectors as follows:

$$\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots \boldsymbol{x}_K\}. \tag{4}$$

The vector $\boldsymbol{x}_i$ in the set represents a salient region or salient object instance of the remote sensing image. Together, these K vectors constitute the region representation of the remote sensing image.

We followed the methodology in [39] to extract the image region features and used a Faster R-CNN [46], a common two-stage object detector. It classifies the category of an object and locates it in a bounding box. The object detection process consisted of two stages. In the first stage, region proposals (with different proportions and sizes) were generated using an RPN, category agnostic scores and candidate boxes were determined using nonlinear maximum suppression, and the candidate box with the highest score was retained. In the second stage, a small-size feature map was obtained after a region of interest (RoI) pooling of the candidate boxes obtained in the first stage. The size of the small feature map depended on the bin size of the RoI pooling. Finally, the small feature map obtained after the pooling was presented to the convolution layer to obtain the category scores and refine the information associated with each proposed box. The region with a supra-threshold intersection over union (IoU) was retained for the model output after nonlinear maximum suppression for each category. Finally, all the regions with supra-threshold detection probabilities were selected. For each selected region i, the region representation was $\boldsymbol{x}_i$ obtained after mean average pooling. Thus, we obtained the vector $\boldsymbol{x}_r$, which represents the analyzed remote sensing image.

We used Faster R-CNN combined with ResNet101 to extract the region features. The ResNet101 is the backbone network that converts input images into feature maps. Since ResNet101 can be divided into C1–C5 blocks, we choose the output of the C4 block and share those feature maps among every region. We then used the C5 block to compute RoI-pooled per-region features. The output of C5 after average pooling constituted the final vector representation of the region features.

C. Mapping Region Features

Region features provide object-level information. However, the RoI pooling operation used in region feature extraction causes information loss, including the loss of detailed information regarding salient objects and object-related background information. Most objects in natural images are evident; therefore, RoI pooling can retain object-related information. However, remote sensing images cover wide areas, and most targets are small, which causes information loss after RoI pooling. To alleviate this information loss, we propose the mapping region features.

When extracting the region features, we define the area in feature map $\boldsymbol{X}$ corresponding to the region feature $\boldsymbol{x}_i$ as $\boldsymbol{X}_{\Omega_i}$. We mask the feature map and filter out the grid features outside the corresponding region to obtain the mapping region feature $\boldsymbol{X}_{\Omega_i}^{\mathrm{mask}}$ for dimensional concordance with the grid features. The entire process is schematically illustrated in Fig. 4. Notably, the grid and mapping region features have the same dimensions

$$\mathrm{Mask}(\boldsymbol{x}(i,j)) = \begin{cases} 1 \times 10^{-6}, \boldsymbol{x}(i,j) \notin \boldsymbol{X}_{\Omega_i} \\ \boldsymbol{x}(i,j), \boldsymbol{x}(i,j) \in \boldsymbol{X}_{\Omega_i} \end{cases}, \quad \boldsymbol{x}(i,j) \in \boldsymbol{X}. \tag{5}$$

D. Fusion Attention Module

To alleviate the information loss due to RoI pooling, we used the mapping region features to compensate for the lost information; however, directly fusing the mapping region and grid features is likely to cause information redundancy and noise. To address this problem, we propose a fusion attention module.

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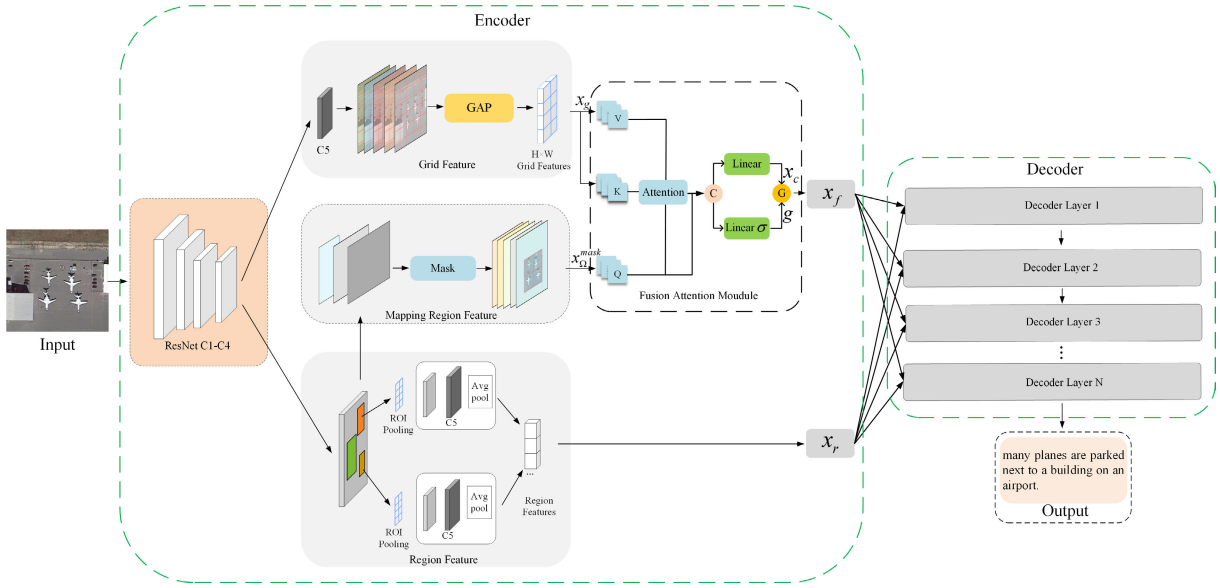


Fig. 3. Structure of the CCT. The CCT consists of an encoder and a decoder; the encoder extracts grid features, region features, and corresponding mapping region features based on the input image and then outputs the fused features $\boldsymbol{x}_f$ and region features $\boldsymbol{x}_r$, which are fed into the cooperative connected decoder to guide the text generation process.

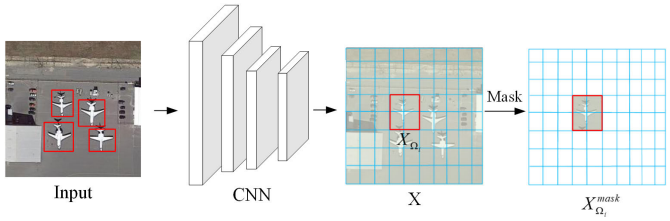


Fig. 4. Mapping region features. The red bounding box in feature map X is $\boldsymbol{X}_{\Omega_i}$, and after the mask operation, the corresponding $\boldsymbol{X}_{\Omega_i}^{\text{mask}}$ is obtained.

We use a multihead self-attention mechanism to construct the relationships between the analyzed features. The multihead self-attention mechanism has two components: 1) a multihead mechanism and 2) a self-attention mechanism. The self-attention mechanism [42] calculates the similarity distribution between vectors $\boldsymbol{Q}$ and $\boldsymbol{K}$ using the dot product and then computes a weighted sum of vector $\boldsymbol{V}$ with the similarity measure between $\boldsymbol{Q}$ and $\boldsymbol{K}$, yielding the following total attention weight:

$$f(\boldsymbol{K}, \boldsymbol{Q}, \boldsymbol{V}) = \text{softmax}\frac{\left(\boldsymbol{Q}\boldsymbol{K}^T\right)}{\sqrt{d}}\boldsymbol{V} \tag{6}$$

where $(d)^{1/2}$ is the scaling factor.

The multihead mechanism expands the ability of the model to attend to different locations. It provides multiple representation subspaces by mapping $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ to h different dimensions, and the attention weight of $\boldsymbol{Q}_i$, $\boldsymbol{K}_i$, and $\boldsymbol{V}_i$ is calculated for each dimension and then concatenated to get the final output. The multihead attention is as follows:

$$f_{\text{multi_att}}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{Concat}(\boldsymbol{h}_1, \boldsymbol{h}_2 \ldots \boldsymbol{h}_H) \tag{7}$$

$$\boldsymbol{h}_i = f\left(\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i\right). \tag{8}$$

The fusion attention module consists of two main parts: a cross feature $\boldsymbol{x}_c$ and a filter gate $\boldsymbol{g}$. The expressions for the cross feature and filter gate are given by the following two linear transformations:

$$\boldsymbol{x}_c = W_q^c \boldsymbol{x}_\Omega^{\text{mask}} + W_v^c \boldsymbol{x} + b^c \tag{9}$$

$$\boldsymbol{g} = \sigma\left(W_q^g \boldsymbol{x}_\Omega^{\text{mask}} + W_v^g \boldsymbol{x} + b^g\right) \tag{10}$$

where W_q^c, W_v^c, W_q^g, and W_v^g are learned matrices, b^c and b^g are the bias matrices, and σ denotes the sigmoid activation function. $\boldsymbol{x}$ is the attention weight between the grid features and mapping region features, and $\boldsymbol{Q}$ corresponds to the mapping region features, while $\boldsymbol{K}$ and $\boldsymbol{V}$ correspond to the grid features, representing the internal relationship between the mapping region features and grid features

$$\boldsymbol{x} = \text{Attention}\left(w_q \boldsymbol{x}_\Omega^{\text{mask}}, w_k \boldsymbol{x}_g, w_v \boldsymbol{x}_g\right). \tag{11}$$

The fusion attention module filters out redundant information and noise by element-wise multiplication of $\boldsymbol{x}_c$ and $\boldsymbol{g}$, yielding the final fusion features as follows:

$$\boldsymbol{x}_f = \boldsymbol{x}_c \odot \boldsymbol{g}. \tag{12}$$

E. Cooperative Connected Decoder

To make the decoder fully perceive different feature types, we propose cooperative connection attention (CCA), which connects the input vector $\boldsymbol{y}$ and different image features $\boldsymbol{x}_f$ and $\boldsymbol{x}_r$ through weighted cross-attention, as shown in Fig. 5. Unlike the Transformer, which only attends to the last layer of the encoder, the cooperative connection enables each decoder layer to attend to different features directly. This improves the matching of image features and the text generated during the text-generation process. Through CCA, the contributions of the different features are summed, and the cooperative connection attention operator is calculated as

$$CC(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{\alpha}_r \odot c(\boldsymbol{x}_r, \boldsymbol{y}) + \boldsymbol{\alpha}_f \odot c\left(\boldsymbol{x}_f, \boldsymbol{y}\right) \tag{13}$$

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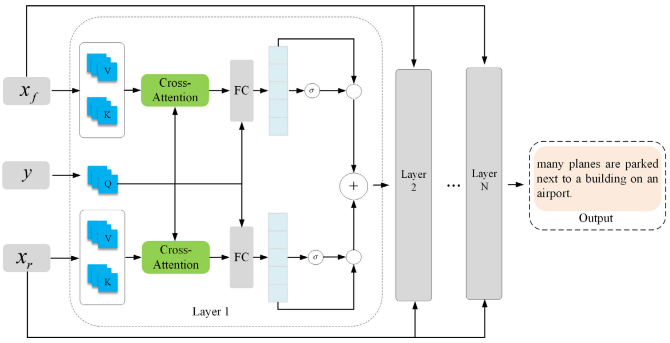


Fig. 5. Cooperative connected decoder. The fusion features $\boldsymbol{x}_f$ and region features $\boldsymbol{x}_r$ are directly connected to each decoder layer; the weight of different features is determined by weighted cross-attention.

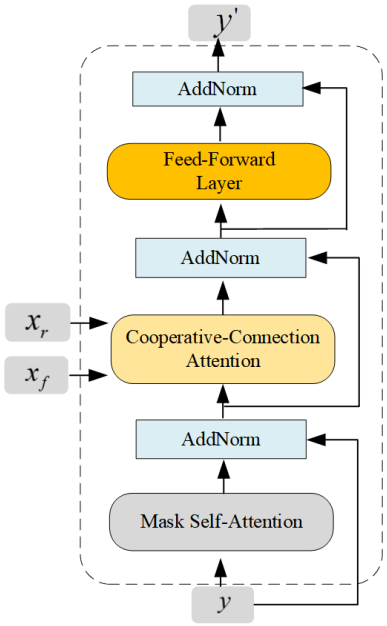


Fig. 6. Structure of the decoder layer. Each decoder layer contains masked self-attention and cooperative connection attention operation; after going through a feedforward layer, the layer's output is obtained.

where $c(\boldsymbol{x}, \boldsymbol{y})$ represents the cross-attention between different features and the input sequence $\boldsymbol{y}$ and $\odot$ denotes the element-wise multiplication

$$c(\boldsymbol{x}, \boldsymbol{y}) = f\left(w_q \boldsymbol{y}, w_k \boldsymbol{x}, w_v \boldsymbol{x}\right), \boldsymbol{x} \in \boldsymbol{x}_r, \boldsymbol{x}_f \tag{14}$$

where w_q, w_k, and w_v are the learned matrices and $\boldsymbol{\alpha}_f$ and $\boldsymbol{\alpha}_r$ are the weight matrices of the fusion features and region features, which determine the contribution and importance of different features, respectively. The weights were obtained as follows:

$$\boldsymbol{\alpha}_r = \sigma(W_r \operatorname{Concat}(\boldsymbol{y}, c(\boldsymbol{x}, \boldsymbol{y})) + b_r) \tag{15}$$

$$\boldsymbol{\alpha}_f = \sigma\left(W_f \operatorname{Concat}(\boldsymbol{y}, c(\boldsymbol{x}, \boldsymbol{y})) + b_f\right) \tag{16}$$

where Concat represents the concatenation operation, whereas b_r and b_f are learnable bias matrices.

Similar to [31] and [42], the cooperative connection decoder is composed of stacked layers, the structure of which is shown in Fig. 6. The predicted words should depend only on generated words. Therefore, the self-attention operation is masked such that the model considers only input element k of step t and the left part of the input sequence. The decoder layer also included a feedforward layer, and each stage required AddNorm operations. Each decoder layer is defined as follows:

$$\boldsymbol{h} = \operatorname{AddNorm}(CC(\boldsymbol{x}, \operatorname{Masked}(\boldsymbol{y}))) \tag{17}$$

$$\boldsymbol{y}' = \operatorname{AddNorm}(ff(\boldsymbol{h})) \tag{18}$$

where $\boldsymbol{y}$ is the input sequence and Masked is the masked self-attention operation. The stacked layers comprise the decoder, which generates the descriptive text. The decoder generally predicts the next word with a probability that depends on the input sequence and current predictions up to the end vector. Thus, a complete descriptive sentence was generated.

V. Experiments

A. Experimental Setup

We used the output of C5 after GAP as grid features to extract the grid features. To extract region features, we pretrained the Faster R-CNN model on the Visual Genome dataset [47] and then trained it on the labeled dataset. After completing the training, we extracted $2048 - d$ image features for each region. The UCM-captions, Sydney-captions, and RSICD were divided according to the division method in [38]. In our experiments, each decoder layer was $512 - d$. We used cross-entropy loss (XE) to pretrain the model. In the pretraining stage, we set the learning rate to 1×10^{-4} to warm up the model during the first five epochs and fixed our learning rate to 5×10^{-7} in the subsequent epochs. We used CIDEr-D as a reward, and in the CIDEr-D optimization stage, the learning rate was 5×10^{-6}. The beam size was 5. The value of the multihead attention parameter h was 8. The batch size was 32.

B. Ablation Experiments

Ablation experiments are conducted to verify the effectiveness of the proposed method. The experimental results are presented in Tables II–VI.

First, we explored the performance of region features in UCM-captions, Sydney-captions, and RSICD. We compared the performances of the region and grid features. For a fair comparison, we used a Transformer as a baseline. We used Faster R-CNN + ResNet101 as the encoder for extracting region features and ResNet101 as the encoder for extracting grid features. After feature extraction, the extracted features were presented separately to the same Transformer decoder. The experimental results are presented in Table II. According to the experimental results, the region features performed similar to the grid features. The performance on each dataset was as follows: region features performed better on the UCM-captions and were comparable to the grid features on Sydney-captions and the RSICD. Experiments have demonstrated that region features can satisfy the requirements of RSIC tasks. However, during text generation, a model based on region features is more "focused" than the one based on grid features (because the RoI is more accurate). This is because, during the model training, the image features learned by the model that uses region features are mainly at the object level

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TABLE II
EXPERIMENTAL RESULTS OF GRID FEATURES AND REGION FEATURES

Dataset	Feature	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE_L	CIDEr
UCM-captions	Grid	0.781	0.721	0.660	0.614	0.357	0.723	2.861
	Region	**0.789**	**0.727**	**0.663**	**0.621**	**0.363**	**0.741**	**2.966**
Sydney-captions	Grid	0.733	0.652	0.581	0.531	0.346	0.682	**2.220**
	Region	**0.741**	**0.656**	**0.590**	**0.533**	**0.351**	**0.691**	2.209
RSICD	Grid	**0.725**	**0.617**	**0.519**	**0.450**	0.346	0.628	2.429
	Region	0.717	0.607	0.516	0.442	**0.363**	**0.635**	**2.431**



Grid:many buildings and green trees are near a pond.

Region:many green trees are around an irregular pond.

(a)



Grid:many buildings are in an industrial area.

Region:many buildings are in two sides of a railway station.

(b)

Fig. 7. Different texts generated by the models that used grid and region features. (a) and (b) Generation results of different images (pond and railway, respectively). The first row shows the sentence generated based on grid features, the second row shows the sentence generated based on region features, and red represents incorrect generation text.

and contain more positional information. The text generated by the model based on region features is more object-related but less related to the overall background, that is, the description is more object-related and lacks an interrelationship between the object and the environment. However, the text generated by a model that uses grid features is more relevant to the overall environment and less attentive to the object itself.

Fig. 7 shows the text generated by the models using different feature types. In Fig. 7(a), the text generated by the model using grid features was focused on buildings and trees near the pond. However, the text generated by the model using region features focused more on the pond. Fig. 7(b) shows that the model with grid features generated incorrect words, and the railway station was incorrectly described as an industrial area. The region feature-based model paid more attention to the central part of the railway station and thus generated the correct text.

1) Mapping Region Features: We explored the role of mapping region features in our model. To eliminate the interference

TABLE III
ABLATION EXPERIMENTS OF MAPPING REGION FEATURES

Dataset	MRF	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE_L	CIDEr
UCM-captions	Region	0.797	0.734	0.682	0.631	0.383	0.679	3.002
	Region + Grid	0.812	0.767	0.711	0.666	0.415	0.709	3.323
	Region + MRF	**0.836**	**0.791**	**0.750**	**0.698**	**0.437**	**0.738**	**3.451**
Sydney-captions	Region	0.788	0.715	0.647	0.597	0.373	0.713	2.311
	Region + Grid	0.800	0.728	0.667	0.612	0.392	0.729	2.457
	Region + MRF	**0.820**	**0.755**	**0.690**	**0.631**	**0.421**	**0.747**	**2.531**
RSICD	Region	0.731	0.623	0.528	0.462	0.353	0.644	2.442
	Region + Grid	0.742	0.627	0.533	0.466	0.355	0.652	2.453
	Region + MRF	**0.752**	**0.633**	**0.541**	**0.471**	**0.361**	**0.657**	**2.461**

TABLE IV
ABLATION EXPERIMENT OF THE FUSION ATTENTION MODULE

Dataset	Fusion module	Mapping region	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE_L	CIDEr
UCM-captions	×	×	0.812	0.767	0.711	0.666	0.415	0.709	3.323
	×	✓	0.853	0.799	0.776	0.713	0.445	0.746	3.563
	✓	✓	**0.922**	**0.890**	**0.864**	**0.833**	**0.573**	**0.883**	**4.156**
Sydney-captions	×	×	0.800	0.728	0.667	0.612	0.392	0.729	2.457
	×	✓	0.823	0.764	0.687	0.634	0.433	0.751	2.542
	✓	✓	**0.866**	**0.799**	**0.747**	**0.691**	**0.460**	**0.791**	**2.867**
RSICD	×	×	0.742	0.627	0.533	0.466	0.355	0.657	2.453
	×	✓	0.756	0.631	0.527	0.460	0.349	0.659	2.449
	✓	✓	**0.798**	**0.693**	**0.608**	**0.533**	**0.383**	**0.692**	**2.881**

of grid features, ablation experiments were conducted using only region features by combining region features with mapping region features and by combining region features with grid features. In the experiments, we separately presented region features, region features + mapping region features, and region features + grid features to the cooperative connected decoder. The experimental results are listed in Table III, where "MRF" represents the mapping region features. After the mapping region features were introduced into the model, all experimental metrics on the three datasets improved compared with the model that only used region features, which was also better than the model with region and grid features.

2) Fusion Attention Module: The effect of the fusion attention module on the performance of the proposed model is explored. We compared the performance of the model with the fusion attention module and a model in which the mapping region features were directly fused with grid features. As shown in Table IV, the experimental results indicated that directly fusing the two feature types did not significantly improve the model's actual performance. After introducing the fusion attention module, all the model metrics improved significantly for the three datasets. This occurred because although the mapping region features added some lost information to the region features, it introduced some redundant information and noise, offsetting the improvement brought about by the mapping region features. After introducing the fusion attention module, redundant information between the mapping region and grid features and the noise generated by the fusion were filtered out. The mapping region features also supplemented the information lost during the extraction of the region features, thereby improving the actual performance of the model. Fig. 8 compares the attention heat maps of the models with and without the fusion attention module

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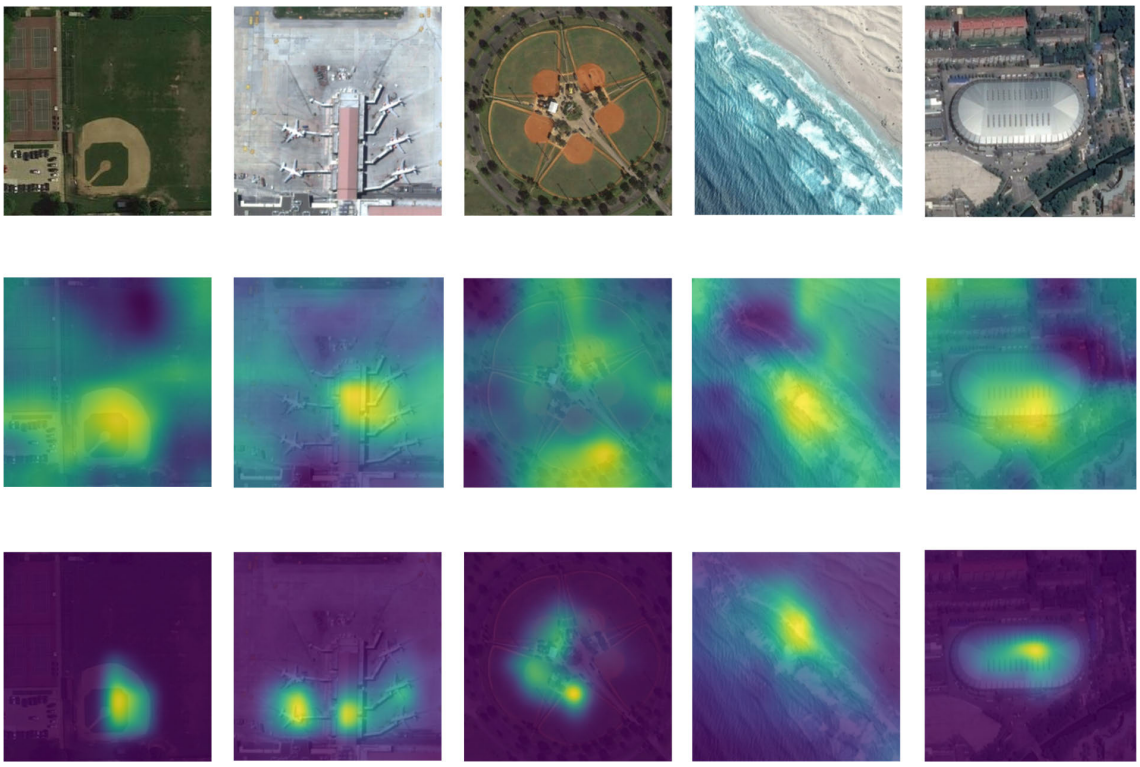


Fig. 8. Attention heatmaps. The first row is the original remote sensing image, the second is the model's attention distribution without the fusion attention module, and the third row is the attention distribution with the fusion attention module. The region with the maximum attention weight in the image is represented in yellow.

TABLE V
EXPERIMENTS OF DIFFERENT GRID FEATURES

Dataset	CGF	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE_L	CIDEr
UCM-captions	GAP	**0.922**	**0.890**	**0.864**	**0.835**	**0.573**	**0.883**	**4.156**
	FC	0.871	0.840	0.813	0.787	0.511	0.819	3.850
Sydney-captions	GAP	**0.866**	**0.799**	**0.747**	**0.691**	**0.460**	**0.791**	**2.867**
	FC	0.833	0.779	0.703	0.641	0.415	0.732	2.714
RSICD	GAP	**0.798**	**0.693**	**0.608**	**0.533**	**0.383**	**0.692**	**2.881**
	FC	0.788	0.676	0.579	0.517	0.379	0.678	2.850

TABLE VI
EXPERIMENTS OF DIFFERENT ATTENTION

Dataset	Connect method	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE_L	CIDEr
UCM-captions	Traditional	0.837	0.789	0.751	0.701	0.433	0.795	3.557
	Cooperative	**0.922**	**0.890**	**0.864**	**0.833**	**0.573**	**0.883**	**4.156**
Sydney-captions	Traditional	0.822	0.755	0.689	0.614	0.413	0.723	2.593
	Cooperative	**0.866**	**0.799**	**0.747**	**0.691**	**0.460**	**0.791**	**2.867**
RSICD	Traditional	0.751	0.644	0.576	0.501	0.371	0.672	2.799
	Cooperative	**0.798**	**0.693**	**0.608**	**0.533**	**0.383**	**0.692**	**2.881**

for the same remote sensing image. After introducing the fusion attention module, the model became more attentive and focused on the main object.

3) Choosing Grid Features: We compared the effects of different grid feature choices when choosing grid features. The experimental results are listed in Table V. We compared the performance of the model that used the fully connected layer output as the grid feature with that of the model that used the GAP output as the grid feature. Here, "CGF" represents the choice of the grid features, "FC" represents the network's output with the fully connected layer, and GAP represents the network's output with the GAP. The experimental results showed that the GAP method performed better than the FC method for both datasets.

4) Cooperative Connection Attention: This study examines the role of cooperative connection attention. We compared the performance of the method using cooperative connections with that of the method using conventional cross-attention. For a fair comparison, in the conventional method, we directly concatenate the region and fusion features as the final output of the encoder and present them to the decoder with cross-attention. The experimental results showed that the decoder with the cooperative connection attention mechanism performed better than the conventional attention mechanisms in the experimental metrics. The experimental results are presented in Table VI. Performance differentiation occurs because the cooperatively connected decoder fully attends to the different features. The decoder attends to both the background information associated with the grid features and the salient region information provided by the region features. Therefore, during text generation, the proposed model can attend to global background-related information and local salient objects. Fig. 9 shows conventional and cooperative connection attention changes during the text generation process. Cooperative connection attention forces the model to focus more on related image areas. In addition, the model focused on the background information associated with the analyzed image. Consequently, the model is more attentive and generates more accurate sentences.

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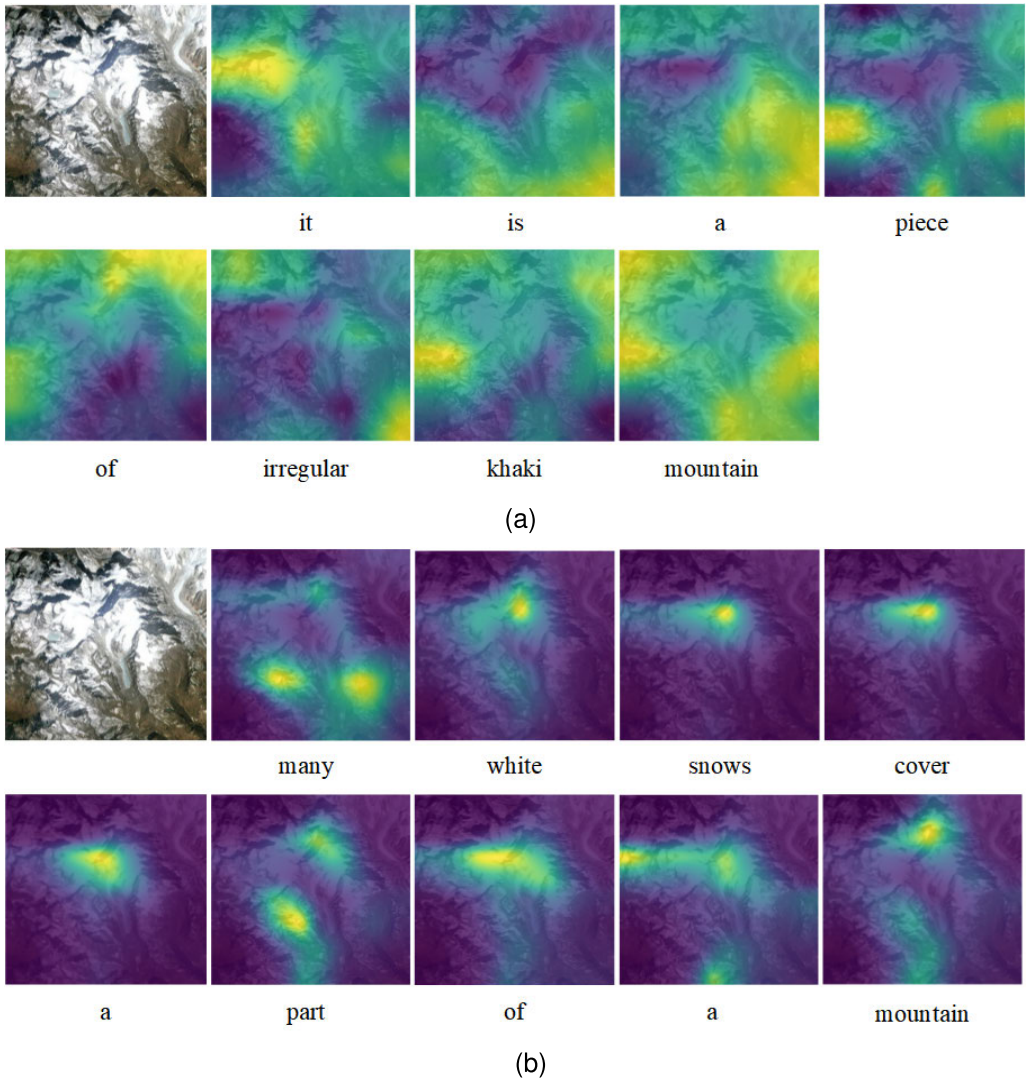


Fig. 9. Text-generation results for different attention models. (a) Attended image region when the transformer generates each descriptive word (cross-attention). (b) Attended image region of our model when generating each description word (cooperative connection attention). The area with maximum attention in the image is represented in yellow.

In summary, we conducted detailed experiments using the proposed method. First, we validated the actual performance of region features in RSIC tasks. Experiments on UCM-captions, Sydney-captions, and RSICD have shown that region features work well in RSIC tasks: region features can provide more object-related information, grid features can provide more background information, and experimental results of mapping region features have shown that the introduction of mapping region features can improve the actual performance of the model on three datasets. Experiments on the fusion attention module have demonstrated that the direct fusion of mapping region features and grid features can improve the model's actual performance to a certain extent, but it introduces noise and redundant information. After introducing the fusion attention module, the noise and redundant information are filtered out, significantly improving the model performance in both metrics. We compared the performance of fully connected layers and the GAP for the selection of grid features. Experiments showed that GAP could provide better global background information than a fully connected layer, and experimental results on cooperative connection attention demonstrated that our method could make the model more sensitive to different features, thus obtaining better experimental metrics.

C. Comparative Experiments

This section compares our method with several state-of-the-art methods, including attention- and Transformer-based methods.

1) *JTTS [17]:* It is a two-stage RSIC model that can be jointly trained. A new operator is proposed to index the multilabel classification results, and an attribute-guided decoder is designed to filter the multilabel prior information obtained by the operator.
2) *DFEN [18]:* An image enhancement module is designed in the encoder, which uses the multiscale characteristics of remote sensing images to realize image enhancement at the image level and the potential visual features of images to guide the text-enhance module in the decoder.
3) *DFT [27]:* It proposes a new DFT structure, which uses the variable dot-product attention mechanism to learn

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TABLE VII
COMPARATIVE EXPERIMENTAL RESULTS ON UCM-CAPTIONS

Dataset	Model	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE_L	CIDEr
UCM-captions	VSCA (2023) [24]	0.890	0.841	0.799	0.758	0.493	0.858	3.831
	MSISA (2023) [20]	0.873	0.810	0.755	0.704	0.465	0.826	3.713
	DFEN (2023) [18]	0.851	0.784	0.728	0.677	0.459	0.805	3.177
	DFT (2023) [27]	0.823	0.770	0.723	0.679	0.444	0.784	3.463
	JTTS (2022) [17]	0.870	0.822	0.779	0.738	0.491	0.836	3.710
	GFT (2022) [29]	0.834	0.772	0.718	0.673	-	0.770	3.315
	MMN (2022) [19]	0.843	0.775	0.711	0.651	0.453	0.785	3.381
	RA (2022) [22]	0.852	0.793	0.743	0.698	0.457	0.807	3.389
	MC(2022) [15]	0.919	0.888	0.860	**0.835**	0.540	0.876	**4.331**
	GLCM (2022) [25]	0.818	0.754	0.699	0.647	0.462	0.752	3.028
	GVFGA (2022) [23]	0.832	0.766	0.710	0.660	0.444	0.785	3.327
	Transformer+ SCST (2020) [28]	0.839	0.790	0.744	0.701	0.446	0.776	3.565
	OUR	**0.922**	**0.890**	**0.864**	0.833	**0.573**	**0.883**	4.156

TABLE VIII
COMPARATIVE EXPERIMENTAL RESULTS ON SYDNEY-CAPTIONS

Dataset	Model	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE_L	CIDEr
Sydney-captions	VSCA (2023) [24]	0.791	0.720	0.661	0.609	0.403	0.735	2.679
	MSISA (2023) [20]	0.764	0.691	0.628	0.573	0.395	0.717	2.812
	DFEN (2023) [18]	0.798	0.697	0.614	0.542	0.373	0.723	2.009
	DFT (2023) [27]	0.837	0.777	0.720	0.666	0.455	0.786	**3.037**
	JTTS (2022) [17]	0.849	0.780	0.714	0.650	0.446	0.766	2.801
	GFT (2022) [29]	0.810	0.732	0.650	0.571	-	0.749	2.567
	MMN (2022) [19]	0.842	0.757	0.672	0.601	0.421	0.733	2.851
	RA (2022) [22]	0.800	0.722	0.653	0.591	0.391	0.721	2.631
	MC (2022) [15]	0.853	0.792	0.734	0.680	0.430	0.742	2.666
	GLCM (2022) [25]	0.804	0.730	0.675	0.626	0.442	0.697	2.434
	GVFGA (2022) [23]	0.768	0.685	0.615	0.550	0.387	0.703	2.452
	Transformer+ SCST (2020) [28]	0.802	0.710	0.631	0.572	0.418	0.726	2.539
	OUR	**0.866**	**0.799**	**0.747**	**0.691**	**0.460**	**0.791**	2.867

the multiscale features of remote sensing images from the foreground and background information.

4) *GFT [29]:* It extracts grid features through pretrained CNN and sends them into Transformer to generate description text. At the same time, self-critical sequence training (SCST) training strategy is used to improve the actual performance of the model.
5) *MMN [19]:* It proposes a two-stage multiscale model structure, in which a multi-interaction feature representation module is proposed. The information between each layer in the module can interact effectively, and the features obtained in different stages can be represented as multiscale features.
6) *RA [22]:* It proposed the RA mechanism to extract high-level attentive maps from encoded features and nonvisual features to help the decoder identify effective information, a semantic gate is proposed in the decoder to achieve a more comprehensive and precise semantic understanding.
7) *Meta Captioning (MC) [15]:* It proposed a meta captioning framework, which uses meta-learning. It extracts meta-features from natural image classification and remote sensing image classification tasks, and the models of the two are trained with class-labeled data so that the models can represent image features from the classification perspective.
8) *VSCA [24]:* It uses a semantic concept extractor to capture semantic concept from remote sensing images and proposes a visual-semantic-co-attention mechanism to realize multimodal interaction of coarse concept-related regions and region-related concepts. Finally, two kinds of attention vectors with semantic-level relational features are incorporated into a consensus-exploitation block to learn cross-modal consensus-aware knowledge.
9) *GVFGA [23]:* It proposes a global vision feature-guided attention mechanism to combine global features with local features to provide image features and proposes a linguistic state guide attention mechanism in decoder to realize the fusion of visual features and text features.
10) *GLCM [25]:* It proposes an attention-based GLCM to obtain the global and local visual feature representation of remote sensing images.
11) *MSISA [20]:* It proposes an MSISA mechanism, which divides the attentive weights into core region, surrounding region, and other regions and focuses those regions differently.
12) *Transformer + SCST [28]:* It propose a new Transformer structure. It added a dropout layer, residual connection, and adaptive feature fusion on Transformer. Reinforcement learning is used to improve the actual performance of the model.

The experimental results are presented in Tables VII–IX. The experimental results showed that the Transformer-based method performed better than the other methods to a certain extent. The best experimental metrics were obtained using the proposed model for most metrics. This was because our model not only used the grid features and retained the background helpful information for subsequent text generation

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TABLE IX
COMPARATIVE EXPERIMENTAL RESULTS ON RSICD

Dataset	Model	Bleu1	Bleu2	Bleu3	Bleu4	METEOR	ROUGE_L	CIDEr
RSICD	VSCA (2023) [24]	0.797	0.686	0.596	0.522	0.375	0.683	**2.934**
	MSISA (2023) [20]	0.784	0.668	0.577	0.504	0.367	0.673	2.844
	DFEN (2023) [18]	0.766	0.636	0.538	0.463	0.373	0.685	2.605
	DFT (2023) [27]	0.758	0.642	0.559	0.492	0.355	0.652	2.581
	JTTS (2022) [17]	0.789	0.680	0.589	0.514	0.377	0.682	2.796
	GFT (2022) [29]	0.774	0.668	0.581	0.510	-	0.678	2.773
	MMN (2022) [19]	0.793	0.681	0.577	0.498	0.374	0.682	2.793
	RA (2022) [22]	0.773	0.665	0.578	0.506	0.363	0.669	2.755
	MC (2022) [15]	0.707	0.594	0.512	0.447	0.339	0.608	2.743
	GLCM (2022) [25]	0.777	0.649	0.564	0.494	0.363	0.678	2.549
	GVFGA (2022) [23]	0.678	0.560	0.478	0.417	0.329	0.593	2.601
	Transformer+SCST (2020) [28]	0.770	0.655	0.563	0.488	0.370	0.670	2.684
	Our	**0.798**	**0.693**	**0.608**	**0.533**	**0.383**	**0.692**	2.881

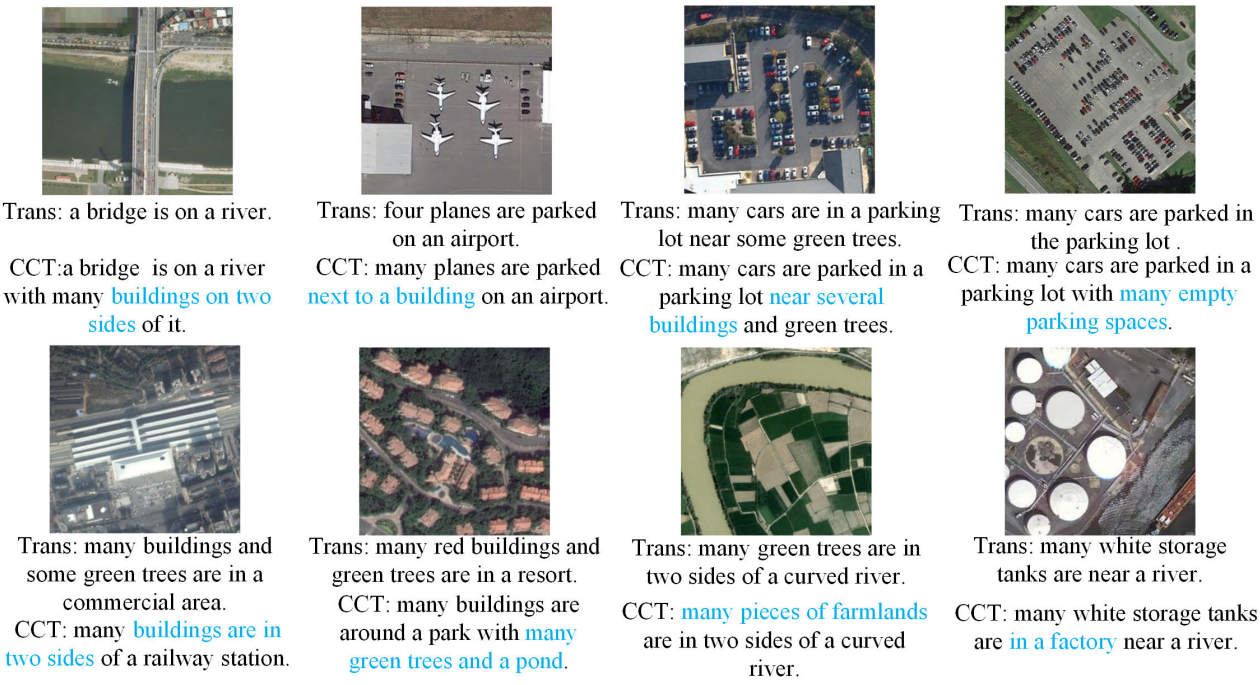


Fig. 10. Quality comparison. "Trans" represents the sentence generated by the Transformer, and "CCT" represents the sentence generated by the proposed model. The blue words represent the surrounding environmental information related to the described object.

but also retained the features of the salient areas or objects in the image, which were improved using the mapping region features. This improves the ability of the model to perceive features in an object's neighborhood. During text generation, the decoder is directly connected to features of all types, unlike conventional Transformer, in which the decoder only attends to the last-layer outputs of the encoder. The configuration used in the proposed method allows different types of features to participate in text generation, providing both global and local area salient information

Quality Comparison: To verify the quality of the generated text, we considered representative results. We chose the transformer as the baseline. The results are shown in Fig. 10. "Trans" denotes the transformer, while "CCT" denotes the proposed model. The blue words represent environmental information related to the described object. By inspecting the generated texts, it is evident that the text generated by our method describes the main object areas in the images, the surrounding environment information, and target-related attributes. In other words, the proposed method can generate descriptive text with richer and more suitable content.

VI. CONCLUSION

As two different representation forms, grid and region features have unique characteristics and advantages. To fully use their advantages and better realize RSIC, we propose CCT for better use of grid and region features in this study. In the CCT, we created a unique design for the grid and region features. We proposed mapping region features to alleviate the information loss problem owing to the extraction of region features and integrated those mapping region features with the grid features using the fusion attention module. To fully utilize the different types of features of the decoder, we propose a cooperative connection attention mechanism in which different types of features are directly connected to each layer of the decoder to guide text generation. To address the problem of insufficient region feature datasets, we proposed standardized annotation principles, supplemented the annotation of region features in

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the RSIC datasets, and extracted region features from this labeled dataset. We validate our model through numerous experiments, and the results demonstrate the effectiveness and superiority of the proposed method. In future studies, we plan to further investigate the use of region features.

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